

Auteur: Siebe Mees
Studierichting: Informatica
Oefeningenreeks 2F nr. 1.2

$$\begin{aligned}\sinh(x - y) &= \sinh x \cosh y - \cosh x \sinh y \\&= \left(\frac{e^x - e^{-x}}{2} \right) \left(\frac{e^y + e^{-y}}{2} \right) - \left(\frac{e^x + e^{-x}}{2} \right) \left(\frac{e^y - e^{-y}}{2} \right) \\&= \frac{(e^x - e^{-x})(e^y + e^{-y})}{4} - \frac{(e^x + e^{-x})(e^y - e^{-y})}{4} \\&= \frac{(e^{x+y} + e^{x-y} - e^{-x+y} - e^{-x-y}) - (e^{x+y} - e^{x-y} + e^{-x+y} - e^{-x-y})}{4} \\&= \frac{2e^{x-y} - 2e^{-x+y}}{4} \\&= \frac{e^{x-y} - e^{-x+y}}{2} \\&= \frac{e^{x-y} - e^{-(x-y)}}{2} \\&= \sinh(x - y)\end{aligned}$$

References